A Survey of Recent Advances in Vision–Language–Action Models: Encoder, VLM, and Action Model Perspectives

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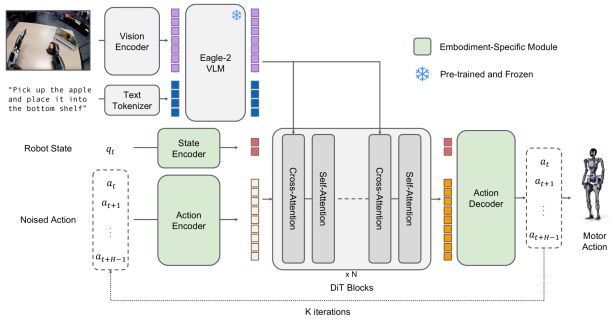


Fig. 1. Overview of a VLA model and its three structural components: the Encoder, VLM, and Action Model. UniT, LaST-R1, and Streaming Flow Policy each target one of these components. Reproduced from [4].

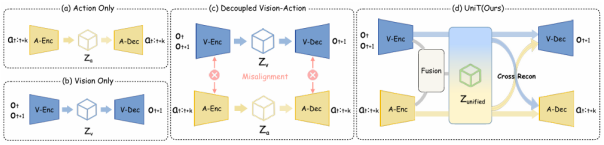


Fig. 2. Comparison of latent action representation paradigms: (a) action-only, (b) vision-only, (c) decoupled vision–action, and (d) UniT's cross-reconstruction. Reproduced from [1].

***Abstract*— Vision–Language–Action (VLA) models have recently emerged as a promising paradigm for general-purpose robotic manipulation, building upon the success of large language and vision–language models. A VLA model can be roughly decomposed into three building blocks: an Encoder that transforms raw observations into compact representations, a Vision–Language Model (VLM) that performs high-level semantic reasoning, and an Action Model that converts intermediate representations into continuous robot actions. In this resume, we review three recent works, each of which targets one of these components. UniT [1] introduces a unified latent action tokenizer that bridges human and humanoid embodiments via visual anchoring, addressing the encoder side. LaST-R1 [2] proposes a reinforcement learning post-training framework that jointly optimizes latent reasoning and action generation, strengthening the VLM. Streaming Flow Policy (SFP) [3] treats action trajectories themselves as flow trajectories to enable low-latency streaming execution, refining the action model. We summarize the background, method, and experimental results of each work, and discuss how their complementary contributions point toward a unified design of practical VLA systems.**

I. INTRODUCTION

Learning manipulation policies that generalize across diverse objects, environments, and language instructions has long been a central challenge in robotics. With the success of large language models (LLMs) and vision–language models (VLMs), the field has rapidly converged on *Vision–Language–Action (VLA) models*, in which images and natural language instructions are mapped to robot actions through a unified neural architecture. Notable examples include OpenVLA [5], π_0 [6], and the GR00T family [4].

Despite their rapid progress, VLA models still face three fundamental difficulties. First, robot demonstration data are scarce by several orders of magnitude compared with the text and video data available to LLMs and VLMs. Second, imitation learning over static datasets struggles with out-of-distribution (OOD) generalization and long-horizon adaptation. Third, diffusion- and flow-matching-based policies, while expressive, suffer from high inference cost and large action latency. These difficulties map closely onto the three structural components of a VLA model, as illustrated in Fig. 1.

The **Encoder** is responsible for extracting embodiment-agnostic representations that link observations and actions. Large pre-trained visual backbones such as DINOv2/v3 [7], [8] and SigLIP [9] have become standard, but the question of *how to learn an action representation* that is robust across embodiments remains open. The **VLM** performs high-level semantic and causal reasoning that determines what action to take. While LLM-style chain-of-thought (CoT) has dramatically expanded reasoning capabilities, the question of how to translate "thinking in language" into "thinking in the body" for closed-loop control is still under active investigation. Finally, the **Action Model** must convert high-level decisions into smooth, continuous trajectories that respect the dynamics of physical robots, balancing expressiveness with real-time controllability.

In this survey, we examine one representative recent paper for each of these components. UniT focuses on the encoder side, learning a unified latent action token space that bridges human and humanoid embodiments. LaST-R1 targets the VLM, introducing reinforcement learning that jointly optimizes latent reasoning and action generation. Streaming Flow Policy redesigns the action model so that flow trajectories coincide with execution trajectories, enabling on-the-fly streaming. We treat these three works as a connected story about pushing VLA systems toward practical general-purpose manipulation.

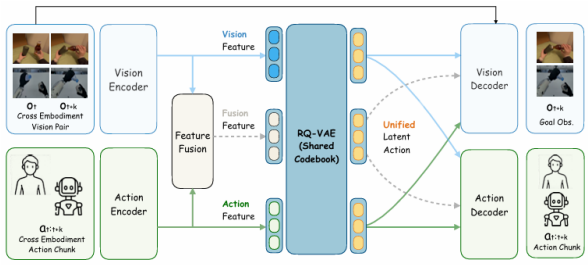


Fig. 3. Architecture of UniT. A tri-branch encoder shares an RQ-VAE codebook, and both decoders reconstruct from every quantized token to enforce cross-reconstruction. Reproduced from [1].

II. Paper 1: UniT — Unified Latent Action Tokenizer via Visual Anchoring

A. Background

A central bottleneck for scaling humanoid VLA models is the scarcity of robot training data. While LLMs consume tens of trillions of tokens, robot demonstration datasets typically contain only millions of trajectories at most. Egocentric human videos offer an attractive alternative because they can be collected at massive scale and contain rich physical interaction priors [10]. However, humans and humanoids differ substantially in degrees of freedom, control modes, and action parameterization, making direct transfer non-trivial.

The traditional remedy is motion retargeting via inverse-kinematics solvers [11], but such pipelines are case-by-case, labor-intensive, and often physically inconsistent. Recent literature has therefore explored *latent action representations* that absorb embodiment differences at the representation level. UniT classifies these methods into three categories and identifies their limitations, as summarized in Fig. 2. Action-only methods [12] are vulnerable to distribution shift because they lack any external visual grounding. Vision-only approaches [13], [14] circumvent the kinematic mismatch but tend to entangle low-level appearance confounders such as lighting and textures, sacrificing motor precision. Decoupled vision–action methods that tokenize the two modalities separately end up with disjoint vocabularies that fail to capture shared physical structure. UniT instead seeks a representation in which vision and action are *mutually reconstructed*, with vision serving as the universal anchor across embodiments.

B. Method

UniT takes as input an observation transition (o_t, o_{t+k}), the current proprioceptive state s_t, and an action chunk $a_{t:t+k}$. As shown in Fig. 3, three parallel transformer encoders summarize this information into modality-specific latents. The *visual branch* consumes frozen DINOv2 features of the observation pair and behaves as an inverse dynamics model, producing a physical-transition latent grounded in the domain-invariant feature space of a powerful visual foundation model. The *action branch* encodes the state and action chunk via embodiment-specific MLPs and a shared transformer to obtain a motor-control latent. The *fusion branch* integrates both modalities and yields a visuo-motor latent that captures complementary structure.

The continuous outputs of the three branches are quantized into a shared discrete codebook $\mathcal{C}$ via a Residual Quantization VAE (RQ-VAE), yielding embodiment-agnostic Unified Latent Action tokens. The defining property of UniT is then *cross-reconstruction*: every quantized token is decoded by both a shared visual decoder, supervised by cosine similarity against the future DINOv2 feature f_{t+k}, and an embodiment-specific action decoder conditioned on s_t that reconstructs the action chunk. The total loss combines reconstruction and quantization terms across all three branches:

$$\mathcal{L} = \sum_{i\in\{v,a,m\}} \Big[\lambda_v \, \mathcal{L}_{\cos}\big(\hat{f}^{(i)}_{t+k}, f_{t+k}\big) + \lambda_a \, \mathcal{L}_{\text{act}}\big(\hat{a}^{(i)}_{t:t+k}, a_{t:t+k}\big)\Big] + \mathcal{L}_{\text{RQ}}. \quad (1)$$

By forcing action features to predict visual outcomes, heterogeneous kinematics are anchored to their physical consequences in pixel space. Conversely, requiring visual features to reconstruct actions strips away appearance noise that has no motor counterpart. What survives this bottleneck is precisely the intersection of the two modalities: an embodiment-agnostic physical intent.

UniT tokens are then deployed in two downstream paradigms. **VLA-UniT** builds on the GR00T n1.5 architecture with a Qwen2.5-VL [15] backbone. The VLM predicts UniT tokens (extracted from the fusion branch) given the current observation and language instruction, and a lightweight flow-matching head subsequently generates embodiment-specific actions conditioned on the VLM features. **WM-UniT** uses the continuous pre-quantization features of the action branch as conditioning signals for the Cosmos Predict 2.5 video generation framework [16], replacing raw action conditioning with embodiment-invariant control tokens.

C. Experiments and Results

UniT is evaluated on the RoboCasa GR1 simulation benchmark covering 24 tabletop tasks, the DROID dataset (95,599 trajectories) [17], and on a real-world IRON-R01-1.11 humanoid with a 50-dimensional action space for Pick & Place and Pouring tasks. Human-to-humanoid transfer is studied via EgoDex.

On RoboCasa GR1, VLA-UniT achieves an average success rate of 66.7% under the full-data setting, outperforming a comprehensive set of baselines including UWM, Diffusion Policy, FLARE, GR00T-N1.6, π-Qwen3, FAST-Qwen3, and OFT-Qwen3. The same architecture without UniT token prediction (GR00T-Qwen2.5) reaches only 47.8%, so the 18.9-point improvement directly quantifies the contribution of UniT as a learning objective. In a few-shot regime with only 10% of the data, VLA-UniT still achieves 45.5%, matching baselines trained on the full dataset. Co-training with EgoDex further boosts in-distribution performance to 50.0% and improves the average OOD success rate from 34.7% to 38.5%.

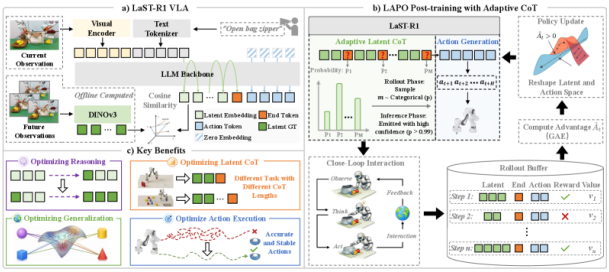


Fig. 4. Overview of LaST-R1. The unified VLA model performs latent reasoning followed by parallel action decoding, and LAPO jointly optimizes both spaces via RL. Reproduced from [2].

In real-world experiments, VLA-UniT reaches 78% on Pick & Place and 75% on the bimanual Pouring task, substantially exceeding the GR00T baseline. Across five OOD categories (Geometry, Distractor, Target, Background, and Combinational), the human-co-trained variant consistently yields the best results. A particularly striking outcome is the unseen Stacking task, where only the human-augmented VLA-UniT attains 60% success, exhibiting emergent upper-body coordination such as waist rotation and head turning. For world modeling, UniT-conditioned generation on DROID outperforms both raw-action and action-only-latent baselines on PSNR, SSIM, LPIPS, FVD, and especially the controllability metric EPE. In a cross-embodiment conditioning study evaluated by Gemini-3-Pro, WM-UniT consistently outperforms raw-action conditioning along semantic, temporal, and geometric dimensions, indicating that UniT tokens carry transferable action semantics across embodiments.

III. Paper 2: LaST-R1 — Reinforcing Robotic Manipulation via Adaptive Physical Latent Reasoning

A. Background

Although VLA models that perform latent reasoning before acting [18], [19] can encode fine-grained physical dynamics, most existing frameworks remain confined to imitation learning over static datasets. This results in compounding errors and limited OOD generalization, particularly in long-horizon and bimanual tasks where small mistakes propagate.

Here, *latent reasoning* refers to deliberation that takes place in a high-dimensional internal feature space rather than in natural language. It is analogous to chain-of-thought (CoT) in LLMs but is tightly coupled with action generation and conveys denser physical information. A parallel line of research has explored online reinforcement learning (RL) for VLA post-training [20], but these methods optimize purely in the action space and bypass the internal reasoning process. LaST-R1 fills this gap by introducing the first RL post-training framework specifically designed for "latent reasoning-before-acting" policies, jointly optimizing the latent reasoning trajectory and the action chunk.

B. Method

The LaST-R1 framework, illustrated in Fig. 4, consists of three tightly integrated components: a unified VLA architecture with hybrid decoding, the Latent-to-Action Policy Optimization (LAPO) algorithm, and an adaptive latent CoT mechanism.

The **VLA architecture** is built on Qwen3-VL-4B [21] with a SigLIP2-Large visual encoder. Given an image and instruction, the LLM backbone first autoregressively produces a sequence of latent reasoning embeddings, terminated by a special `<latent_end>` token. Continuous actions are then discretized via a parameter-free tokenizer and produced *in parallel* from a single forward pass with bidirectional attention, which substantially improves inference efficiency. The latent reasoning targets are derived from the $\langle\text{CLS}\rangle$ token of DINOv3 with channel-wise top-k selection ($k = 2560$); these targets are precomputed offline so that no extra inference cost is introduced. A value head shares the backbone for advantage estimation.

The core contribution is **LAPO**, a PPO-style RL algorithm that treats latent embeddings as implicit decision variables. For the discrete action sequence at step t, the standard categorical ratio $r_t^a(\theta) = \exp\big(\log \pi_\theta(C_t) - \log \pi_{\theta_{\text{old}}}(C_t)\big)$ is used. For the continuous latent reasoning embeddings, LAPO approximates the rollout distribution by an isotropic Gaussian centered at the current policy output, which yields a closed-form step-level latent ratio:

$$r_t^z(\theta) = \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{N_z} \left\| z_{t,i}^{\text{old}} - z_{t,i}^{\theta} \right\|^2 \right). \tag{2}$$

The clipped surrogate loss is then formed jointly over $\{r_t^a, r_t^z\}$ with advantages computed by Generalized Advantage Estimation. When $\hat{A}_t > 0$, the optimization explicitly pulls the current policy's latent embeddings toward the "good-reasoning" manifold that produced successful trajectories, while simultaneously shaping the action space. The total objective combines action loss, latent loss, and value loss with weights λ_1 and λ_2.

Finally, the **adaptive latent CoT** mechanism allows the model to vary the reasoning horizon per task. Within a maximum length $N_{\max} = 8$, the `<latent_end>` token is allowed to be emitted only at M predefined candidate positions. During inference, the model exits as soon as it predicts `<latent_end>` with probability $p \geq 0.99$. During RL rollouts, the candidate lengths are sampled from a temperature-controlled categorical distribution

$$p_m = \frac{\exp(l_m/\beta)}{\sum_{i=1}^{M} \exp(l_i/\beta)}, \tag{3}$$

and a separate transition-token loss $\mathcal{L}_{\text{end}}$ is added with weight λ_3, encouraging short reasoning on easy states and longer deliberation on harder ones. The ablation study identifies $M = 4$ as the best trade-off between flexibility and stability.

C. Experiments and Results

In simulation, LaST-R1 is evaluated on the LIBERO benchmark [22], which consists of four task suites (Spatial,

Object, Goal, Long) with 10 tasks each. Under a particularly demanding data regime—*one-shot* SFT warm-up followed by online RL—LaST-R1 attains an average success rate of 99.9%, surpassing OpenVLA, π_0, $\pi_{0.5}$, OpenVLA-OFT, SimpleVLA-RL, RLinf-GRPO, and πRL. The advantage is most pronounced on LIBERO-Long, where LaST-R1 reaches 99.8% compared to 94.0% for the next-best RL baseline, demonstrating the benefit of latent CoT for long-horizon planning. Learning-curve analysis shows that LAPO converges substantially faster than action-only PPO and avoids the saturation observed for the baseline on Long-horizon tasks.

In real-world experiments on a Franka Research 3 platform, LaST-R1 is evaluated on one single-arm task (hexagon block insertion) and three dual-arm tasks (opening a bag zipper, wiping a vase with a sponge, opening a bottle cap). With only 30 trajectories per task for SFT warm-up followed by human-in-the-loop RL (updating LoRA parameters of rank 32), the average success rate rises from 52.5% to 93.75%, exceeding the full-SFT $\pi_{0.5}$ baseline (71.25%) by 22.5 points. Under severe perturbations such as unseen objects, background changes, and lighting variations, $\pi_{0.5}$ degrades by up to 45 points, while LaST-R1 retains performance within 15 points and shows nearly no degradation against background or lighting changes.

Further analysis reveals that after LAPO post-training, the reasoning-length distribution sharply concentrates on short horizons (lengths 2 and 4), and the execution episode lengths drop below those of the simulator-provided expert demonstrations on most task suites. This indicates that the optimized latent reasoning process not only matches but exceeds the temporal efficiency of scripted experts, while remaining robust to OOD conditions.

IV. Paper 3: Streaming Flow Policy — Treating Action Trajectories as Flow Trajectories

A. Background

Diffusion- and flow-matching-based policies [23] have become a popular choice for behavior cloning because they can represent multi-modal action distributions. This is important because for many tasks (e.g., picking up a cup), multiple distinct trajectories are valid, and policies trained with MSE losses tend to average them into degenerate behaviors.

However, conventional diffusion/flow policies operate in the *action-sequence* space $\mathcal{A}^T$ and start sampling from pure Gaussian noise. As a result, they sample a "trajectory of trajectories" and incur two main drawbacks: high sampling cost from discarding all intermediate trajectories, and inability to stream actions to the robot until the full denoising loop completes. Streaming Diffusion Policy [24] partially mitigates the latency issue but requires maintaining a rolling buffer with increasing noise levels across the entire prediction horizon. Streaming Flow Policy (SFP) takes a more fundamental approach: it treats the action trajectory itself as a continuous normalizing flow trajectory in action space, aligning "flow time" with "execution time" so that actions can be streamed on the fly.

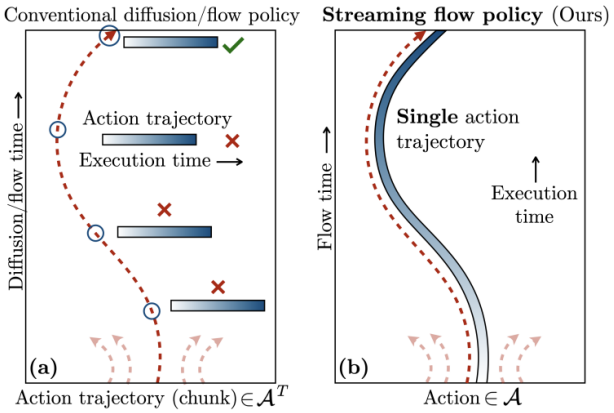


Fig. 5. Conventional diffusion/flow policies (left) sample a "trajectory of trajectories" in $\mathcal{A}^T$, while Streaming Flow Policy (right) treats the action trajectory itself as a flow trajectory in $\mathcal{A}$, enabling on-the-fly streaming. Reproduced from [3].

B. Method

SFP models the policy as a history-conditioned velocity field $v_\theta(a, t \mid h)$ on the action space $\mathcal{A}$, parameterized by a neural network. Unlike conventional flow policies, the initial action a_0 is sampled from a narrow Gaussian centered at the most recently executed action a_{prev}, i.e., $a_0 \sim \mathcal{N}(a_{\text{prev}}, \sigma_0^2)$. Flow time $t \in [0, 1]$ is aligned with the (rescaled) execution time of the action chunk, as illustrated in Fig. 5.

Given a demonstration trajectory ξ, SFP analytically constructs a *stabilizing* conditional velocity field:

$$v_\xi(a, t) = \dot{\xi}(t) - k\,\big(a - \xi(t)\big), \tag{4}$$

where the first term is the trajectory velocity along ξ and the second term is a proportional stabilization term that pulls deviations back toward the demonstration. The induced per-timestep marginal distribution is a Gaussian tube around ξ whose standard deviation decays exponentially over time:

$$p_\xi(a \mid t) = \mathcal{N}\big(a \mid \xi(t),\, \sigma_0^2 e^{-2kt}\big). \tag{5}$$

This stabilization is motivated by prior theoretical results on imitation learning showing that low-level stabilizing controllers can reduce distribution shift and tighten generalization bounds.

The marginal velocity field is then trained via flow matching, with $v_\xi(a, t)$ as the regression target:

$$\mathcal{L}_{\text{CFM}}(v, p_{\mathcal{D}}) = \mathbb{E}_{(h,\xi)\sim p_{\mathcal{D}}} \mathbb{E}_{t\sim U[0,1]} \mathbb{E}_{a\sim p_\xi(a|t)} \big\| v(a, t \mid h) - v_\xi(a, t) \big\|_2^2. \tag{6}$$

The flow-matching theorem guarantees that the minimizer's marginal action distribution at every t matches the training distribution, preserving multi-modality even when many demonstrations are aggregated.

At inference time, σ_0 is set to zero so that the policy behaves deterministically. An ODE solver autoregressively integrates the velocity field, and each newly computed action can be streamed directly to the robot controller without waiting for the full chunk to complete. When the action

space corresponds to robot configurations tracked by a stiff low-level controller, SFP can be used for *state imitation* by initializing integration from the measured robot state, leveraging proprioception to reduce accumulated error.

SFP comes with one principled caveat: while it matches the marginal action distribution at every timestep, it does not necessarily match the joint distribution over entire trajectories. The authors argue that for most quasi-static manipulation tasks this is a desirable feature, since it permits the policy to compose new valid behaviors from segments of training demonstrations, and they prove that local constraints (joint limits, convex velocity constraints) are still respected.

C. Experiments and Results

SFP is evaluated in simulation on the Push-T environment [23] with both state and image inputs, and on the RoboMimic Lift, Can, and Square tasks. Baselines include Diffusion Policy with 100 DDPM steps, Diffusion Policy with 10 DDIM steps, conventional flow-matching policy, and Streaming Diffusion Policy. An ablation with $k = 0$ measures the contribution of the stabilization term.

On Push-T with state inputs, SFP attains an average/maximum state-imitation score of $95.1\%/96.0\%$, outperforming DDPM-100 ($92.9\%/94.4\%$) and all other baselines. The action-imitation setting yields $91.7\%/93.7\%$, again best among all methods. On RoboMimic, SFP achieves 100% on Lift, $98.4\%/100\%$ on Can, and $78.0\%/84.0\%$ on Square, matching or exceeding every baseline. Crucially, SFP achieves an action latency of 4.5 ms, an order of magnitude lower than DDPM-100 (53.4 ms) and several times faster than Streaming Diffusion Policy. The ablation with $k = 0$ drops the Square success rate from 78.0% to 53.2%, confirming that the stabilization term is essential for mitigating distribution shift.

Real-world experiments on a Franka Research 3 arm with a RealSense D435f camera—reaching and picking an apple, and reorienting a block—show that SFP produces noticeably smoother and faster motions than Diffusion Policy. Importantly, this advantage stems not just from lower mean latency but from the fact that actions are emitted at a uniform interval as the integration proceeds. In contrast, diffusion policies must complete the entire denoising loop before any action is sent, which yields a stop-and-go execution pattern even when the average latency appears similar.

V. CONCLUSIONS

We have surveyed three recent papers that target distinct components of the VLA model. UniT addresses the **Encoder** side by introducing a visually-anchored, cross-reconstruction-based tokenizer that bridges human and humanoid embodiments into a shared physical language. This token space induces alignment not only at the input level but also within the internal representations of downstream VLA policies and world models, enabling data-efficient policy learning, robust OOD generalization, and even zero-shot task transfer from human demonstrations. LaST-R1 strengthens the **VLM** component by introducing LAPO, a reinforcement learning algorithm that jointly optimizes latent reasoning and action generation, combined with an adaptive latent CoT mechanism. With only one-shot supervised warm-up, the resulting policy attains near-perfect success on LIBERO and substantially outperforms strong SFT baselines on contact-rich real-world bimanual tasks, demonstrating the value of importing test-time scaling ideas from LLMs into robot control. Streaming Flow Policy reshapes the **Action Model** by treating action trajectories themselves as flow trajectories, achieving multi-modal but streamable execution with an order-of-magnitude speedup over diffusion-based policies, and producing visibly smoother behavior on physical robots.

Although these three works appear to address independent problems, they share a common direction: making VLA systems closer to practical, general-purpose robotic foundation models. The embodiment-agnostic representation of UniT, the reasoning-augmented post-training of LaST-R1, and the real-time streaming generation of SFP are mutually compatible and arguably complementary. One can envision a future VLA model in which UniT-style tokens serve as the latent reasoning target predicted by a LaST-R1-style VLM, and the resulting context features condition an SFP-style flow head that streams actions to the robot in real time. Such an integrated architecture would combine *cross-embodiment scalability*, *adaptive deliberation*, and *reactive control* within a single system, bringing us closer to general-purpose robotic intelligence that learns from massive heterogeneous data, reasons over diverse physical situations, and acts smoothly in the real world.

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